

Review

Management of water repellency in Australia, and risks associated with preferential flow, pesticide concentration and leaching

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Abstract

The three most westerly states of southern Australia have the largest area of water repellent soils, which limit agricultural production, of any country in the world.

Simplified principles of the problems caused by repellency and the principles of soil management solutions are considered and related to experimental evidence. The phenomena of diverted soil water flow and isolated dry soil can explain most of the problems caused by repellency. Plant adaptation, soil or hydrophobic removal, reduced soil drying, reduced surface tension, water harvesting, avoidance, masking and, perhaps, water movement along dead root systems are the main soil management principles.

Dead roots may play a role in zero till cropping systems, allowing more uniform wetting of dry hydrophobic soil at the base of a dead plant and along the dendritic pattern of the dead root system.

Application of these management principles, especially water harvesting, avoidance and masking (by the use of deep trenching, furrow sowing methods or claying), have made a considerable improvement to sustainability and productivity of farming systems on the water repellent soils of Australia.

Evidence is selected to assess risks of preferential flow, pesticide concentration and leaching for different agricultural soil management methods. All management methods can have some risks, but claying seems to have the least risk and furrowing the highest risk of encouraging preferential flow, pesticide concentration and leaching.

It is suggested we have insufficient information and understanding to quantify the risks of groundwater contamination for different environments, farming systems and soil management methods to control repellency.

There is an urgent need to develop quantified guidelines to minimise any possible groundwater contamination hazard for the extensive areas using farming systems with furrows and increasing amounts of pesticide and fertiliser. © 2000 Elsevier Science B.V. All rights reserved.

Keywords: Water repellence; Soil management; Preferential flow; Leaching; Pollution; Herbicides; Cropping systems; Modelling

1. Introduction

The three most westerly states of southern Australia have the largest area of water repellent soils, with limitation to agricultural animal and grain production, of any country in the world. About 2 Mha in Western Australia, 2 Mha in South Australia and 1 Mha in Victoria are known to be affected and a further large

area has the potential to be repellent, due to soil textures associated with low surface area (i.e. a further 3 Mha in WA; Blackwell, 1993; Moore and Blackwell, 1998). Thus in Australia an area of at least 5 Mha, equivalent to a larger area than the Netherlands (4.8 Mha), has agricultural production affected by water repellence.

Australia is a very dry continent with peripheral

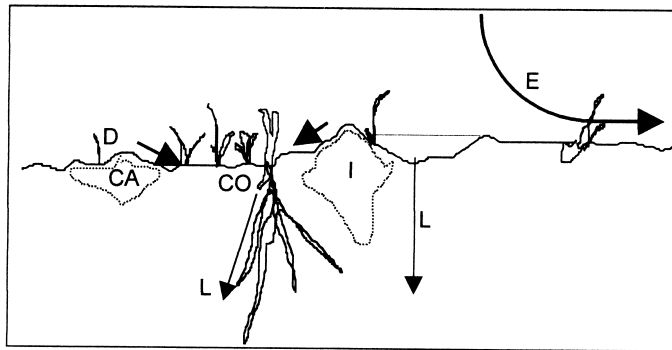


Fig. 1. A schematic outline of processes encouraged in cropping water-repellent soils without improved management; the letters are explained in the text.

farming areas dependent on intercepted precipitation and groundwater for potable water supply. In WA surface water dams and bores from perched groundwater are the main source of potable water supply in rural areas. Farming practices can influence the quality of groundwater. Hirshberg and Appleyard (1996) reported an increase in ammonium and nitrate concentrations and high sulphate/chloride ratios as a result of widespread fertiliser use in farming areas north of Perth.

Application of soil management principles to overcome or use repellency, especially water harvesting, avoidance and masking (by the use of furrow sowing methods, deep trenches, or claying) have made a considerable improvement to sustainability and productivity of farming systems on the water repellent soils of Australia in recent years. However, there are still concerns about accelerated groundwater contamination from preferential flow and leaching of pesticides and nutrients through water repellent soils.

A current cause for concern is the increasing use of the pesticide atrazine in the cultivation of triazine tolerant (TT) Canola. Western Australia's 1998 TT canola crop is estimated at 350,000 ha. A large proportion of this area is on water repellent soil and much of this is sown dry in northern areas, due to the long growing season needed for the varieties used. The Australian National Registration Authority for Agriculture and Veterinary Chemicals (NRA) has only allowed a temporary permit for atrazine use in WA for 1998; limited to two applications per season and a maximum of 4 L/ha for soils with pH < 6.5 and 2 L/ha for soils with pH > 6.5. A monitoring program

is also being conducted by Agriculture Western Australia to comply with the temporary permit. The growing of high yielding wheats with more resistance to leaf diseases on water repellent soils is also encouraging more use of aphicides and fungicides in some higher rainfall areas. This may also be a cause for concern.

Management of water repellency in Australia is very challenging if groundwater contamination risks are to be reduced. The extensive occurrence of the problem and small financial margins of many agricultural enterprises often constrict management methods to low cost and simplicity; furrow sowing methods are in this category. The most effective solution to repellency, and the management method with potentially the lowest risk of groundwater contamination is probably claying; because it can remove the water-repellent behaviour of the soil. Where appropriate clay is available and the financial return over a few years is encouraging, the process of claying is attractive; but it is still relatively slow, expressed as area treated per year. Therefore, it is not advisable to rely on claying as the only solution to groundwater contamination by crop husbandry practices. It would be better to devise management guidelines and simplified decision aids which are based on knowledge of the principles involved and evaluation of the risks by quantitative modelling of leaching. This paper outlines the principles involved in repellency problems and solutions as well as presenting some evidence for specific processes contributing to preferential flow, pesticide concentration and leaching. Suggestions are then made for further research and development for new

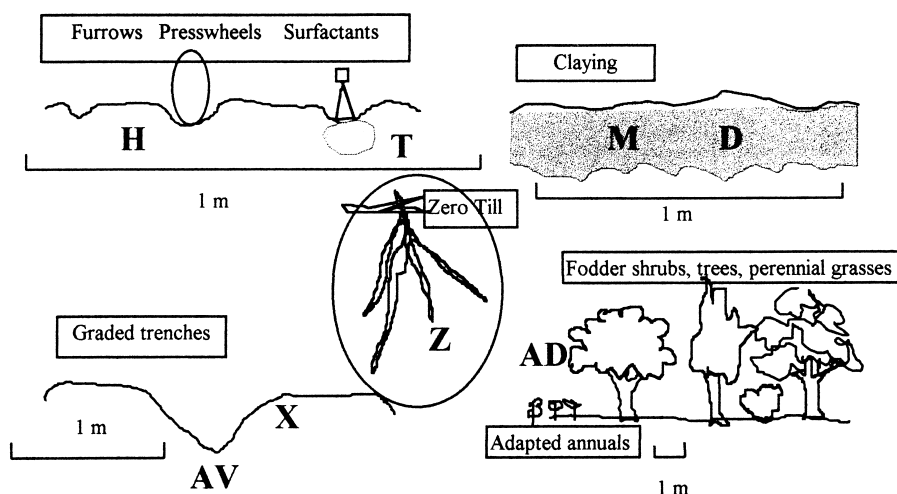


Fig. 2. A schematic outline of processes used in management of water repellency, letters are explained in the text. The relative scale is shown by bars, furrow sowing, claying and zero till share the same scale.

management tools to minimise groundwater contamination risk due to cropping water repellent soils.

2. Principles of water repellency problems

Simplified principles of the problems caused by repellency and the principles of soil management solutions can be considered to provide a conceptual framework for discussion of groundwater contamination problems within management solutions.

Water repellency of dry soil can lead to the following problems for agricultural production: (letters relate to Fig. 1).

- More water erosion and wind erosion, due to increased lateral flow and poor cover (E).
- Isolated dry soil and poor topsoil wetting, due to preferential vertical infiltration (I).
- Accelerated leaching of solutes due to preferential water movement (L).
- Delayed emergence of weed and crop plants from the initially unwetted zones (D).
- Concentration of some solutes (e.g. pesticides) by lateral flow and splash (CO).
- Carry-over of seeds, unactivated pesticides and fertilisers in isolate dry soil (CA).

Staggered germination of weeds can create complications for weed control by selective pesticides which

are more efficient when the weed and crop are at predictable and uniform growth stages.

An extreme example of preferential flow is the observation of water moving 'through' dry soil zones along the dry remains of plant roots on excavation of soil beneath infiltrometers. Emmerson (1954) reported laboratory and field evidence of preferential water movement through severed roots of grasses below the depth of autumn cultivation of an English clay soil. He partially attributed this to an assumed hydrophilic nature of the central meta-xylem cavity of species such as *Lolium perenium*, along which the water could move as a syphon in laboratory experiments. A consequence of such preferential movement of water along dead roots may be the observation of early wetting of old plant rows by the first rains of the following season on water repellent sands in Western Australia. Dekker and Ritsema (1997) have made similar observations of preferential soil wetting along dead maize roots which grew in water repellent soil. In extreme cases of isolated dry soil volumes, crop seed planted in one year has been observed to germinate in the following year, after another crop has been sown.

In contrast to agricultural production problems, it is possible that preferential flow to established plants and seeds of competitive plants isolated in dry zones may be an advantage to the survival of native vegetation. Such vegetation often grows on deep-water

repellent siliceous sands with poor water holding capacity and suffers much competition for water and nutrients.

The phenomena of diverted soil water flow, both laterally and vertically, and isolated dry soil can explain most of the problems listed above caused by repellency. Accelerated leaching, erosion and concentration of pesticides is a consequence of diverted flow. Poor plant establishment and cover, as well as carry-over of pesticides and fertilisers is a consequence of isolated volumes of dry soil.

3. Principles of water repellency solutions

Management options for water repellent soils used for agriculture are quite numerous, compared to many other soil problems, such as acidity. The various options can be explained in terms of simplified principles (Blackwell, 1993). These are explained below, the letters relate to the schematic diagram in Fig. 2.

3.1. Adaptation

Perhaps the oldest solution in Western Australia was to select pasture species which were naturally adapted to water repellent sands (AD); Blue Lupins (*Lupinus cosentinii*) are used as animal pasture and are able to germinate and establish on the soil surface, thus avoiding any need for burial and being isolated in dry water repellent soil. Fodder shrubs and trees, once established, rely more on the subsoil than topsoil for water supply and nutrition, thus they have also been adapted to avoid some repellency problems and provide animal feed and timber for agricultural enterprises.

3.2. Avoidance

Avoidance of water repellent topsoil (AV) is also a well established practice in some areas, especially for establishment of trees and fodder shrubs. Avoidance involves removal the topsoil by lateral grading of a deep trench. Seeds or seedlings are then planted in the non-repellent subsoil. Some forms of furrow sowing allow this for crop establishment, by sowing the crop seed onto shallow moist soil below dry surface soil.

3.3. Water harvesting

Furrow sowing of various kinds employs water harvesting (H) to divert the preferential flow to the seed. Various combinations of points, discs and press-wheels are used to make furrows.

3.4. Reduced surface tension

Surfactant bands in furrows, to make more economical use of surfactants, employ reduced surface tension (T) to enhance soil wetting.

3.5. Reduced drying

Zero till methods in general can reduce soil drying by accumulation of crop residues to minimise soil drying and the development of repellency (Z) and may employ dead root systems from previous crops and pastures to assist redistribution of water during infiltration.

3.6. Masking and degradation

Claying initially employs soil particle masking (M) to cover the hydrophobic surfaces, and may encourage degradation of hydrophobic material (D) by improved microbiological activity. Recent experiments by Roper (1998) show relatively small amounts of clay and lime added to water repellent soil can encourage microbial degradation of hydrophobic material.

3.7. Export

In extreme cases, repellent soil can be removed and exported (X) to other locations in the profile or local area. Such practice is not common in Australia due to the high erosion risks created by such soil exposure. This happens on a small scale during soil trenching.

4. Common principles affecting the risk of groundwater contamination from repellency problems and management solutions

The most obvious principle or process increasing risk of groundwater contamination from cropping water repellent soil is preferential flow, either across the surface, through hollows or furrows and along unstable wetting fronts.

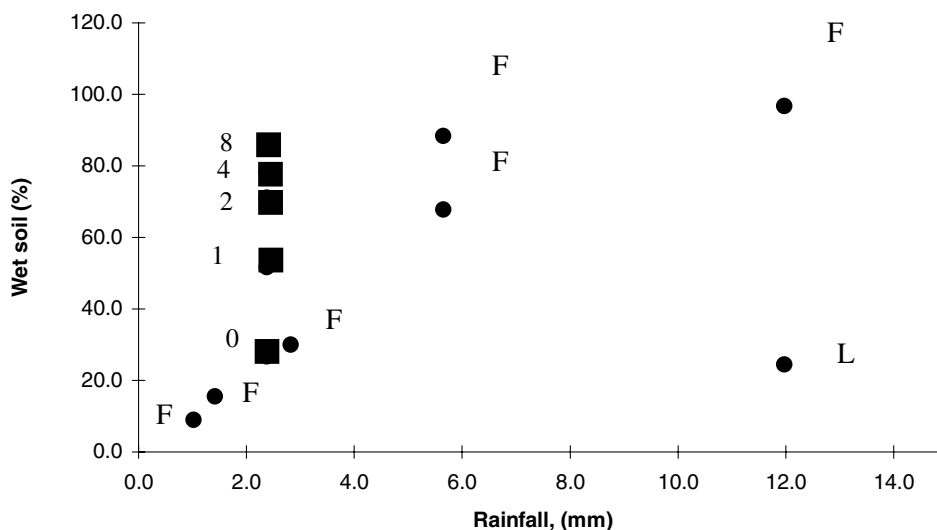


Fig. 3. Soil wetting 40 mm below the bottom of furrows (F) or a level surface (L). Wetting after addition of banded wetting agent previous to the rain is also shown (■) and the rate of application, on a total area basis, indicated in L/ha.

This occurs both in unimproved soil (CO and L in Fig. 1), as well as in improved systems using water harvesting (H in Fig. 2).

Principles and processes which may reduce the risk of groundwater contamination are:

- Masking from claying, which can reduce repellence and increase adsorption of solutes. Dellar et al. (1994) showed that phosphorous retention of water repellent sand was increased by incorporation of clay. Despite this increase in the ability to retain nutrients by claying, the measured values by Dellar et al. (1994) are still in the range of values of Phosphorous Retention Index (PRI) of <2. Weaver and Summers (1998) classified such values of PRI as still having the greatest risk of losing phosphorous by leaching.
- Increased macro-organic matter accumulation by zero-tillage practices. Although increased organic matter can generally increase repellency, in a frequently cultivated system, zero till offers the benefits of reduced soil drying by surface residues restricting evaporation. Such conditions may improve the microbial environment, with amelioration of nutritional limitations. More organic matter and microbial activity may increase adsorption and degradation of pesticides. Water movement along dead root systems may enhance such interaction.

- Reduced surface tension from surfactants may increase adsorption of some nutrients and pesticides by surface organic matter, due to the greater wetted volume of soil available for adsorption when surfactant is used; see Fig. 3.
- Isolated dry soil can hold some nutrients and pesticides which may be released when and if the soil is eventually wetted. Patches of unwetted soil occur in unameliorated conditions and after some forms of furrow sowing. Delayed release of solutes from initially dry zones may reduce the concentration in a 'slug' of leached pesticide and nutrient.

5. Relative occurrence of management solutions

The most extensive management solutions for water repellency currently used in Australia are for improved cropping. Furrow sowing, usually as part of a no-till system, using narrow points and press-wheels is the currently most extensive method used in cropping. Claying is becoming more common, especially in the south coast of Western Australia, South Australia and Victoria. Large plantations of fodder shrubs can be found in the water repellent soils north of Perth. Surfactants are used commonly in the establishment of fodder from seed and are becoming more common in some parts of Western

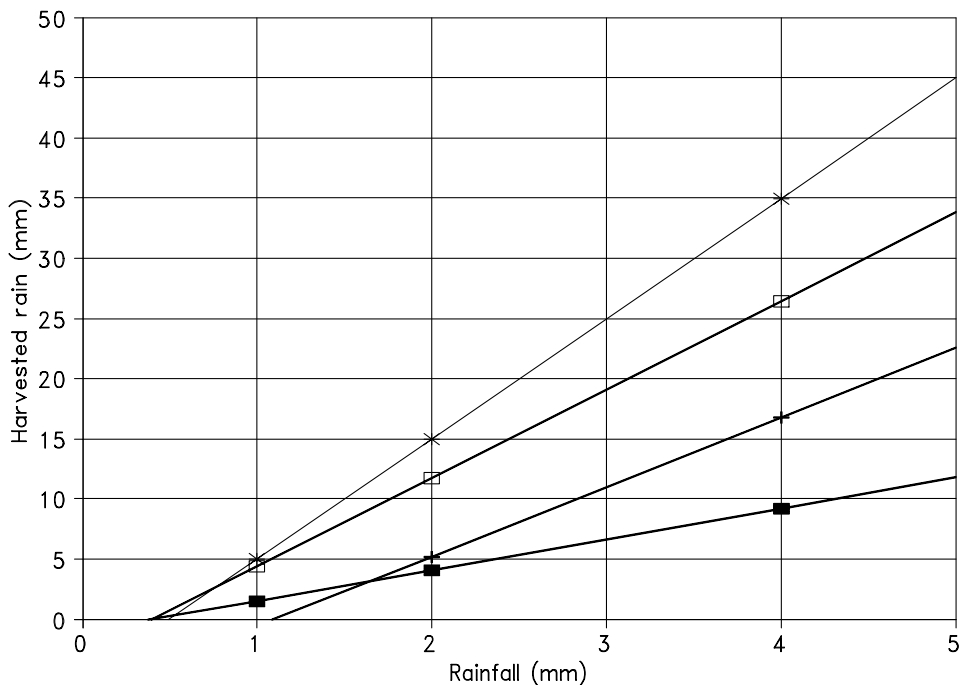


Fig. 4. Results of a simple model of water harvesting#. The influence of a bare surface (X) prostrate stubble (+) or erect stubble (□) is shown for a spacing of 500 mm, as well as erect stubble for a spacing of 200 mm (■). 50 mm/h is the assumed rainfall rate. # $H = ((P - I)(S - C)R)/C$; where H = harvested water (mm); P = rainfall (mm); I = rain to initiate runoff (mm); S = spacing or furrow width (mm); C = collection width in base of furrow (mm); R = runoff ratio.

Australia and South Australia for crop establishment, in conjunction with furrows. Zero till methods, using full stubble retention and crop seeding by precision disc machines, are growing in use. Thus, the most extensively used management principle for management of water repellency in Australia is water harvesting, which is essentially the control of a process which is a common problem feature of water repellent soils, preferential or diverted flow.

6. Choice of management solutions

Current advice for choices of soil management methods to reduce repellence problems are based on the current repellency status of the soil, the risk of the soil developing repellency, the type of farming system being used as well as the availability of clay and the ability to finance a more expensive solution from the short term profits of a less expensive one (Moore and Blackwell, 1998). Abadi (1994) has outlined a method

for choosing the relative investment in different repellency management solutions, depending on the amount of repellent soil and the availability of different solutions. The addition of management guidelines to minimise groundwater contamination risks may also be able to be added to such decision aids in the future.

7. Risks of preferential flow

Water harvesting into furrows can be an advantage for small rainfall events (>about 4 mm) and in dryer seasons (Blackwell, 1997). As the rainfall events become larger (approx. >50 mm), of increased intensity and the season becomes wetter, the advantages of water harvesting can become a disadvantage of preferential water flow, leading to furrow filling and leaching. In this section some experimental evidence of preferential flow through water repellent soils in Western Australia is shown.

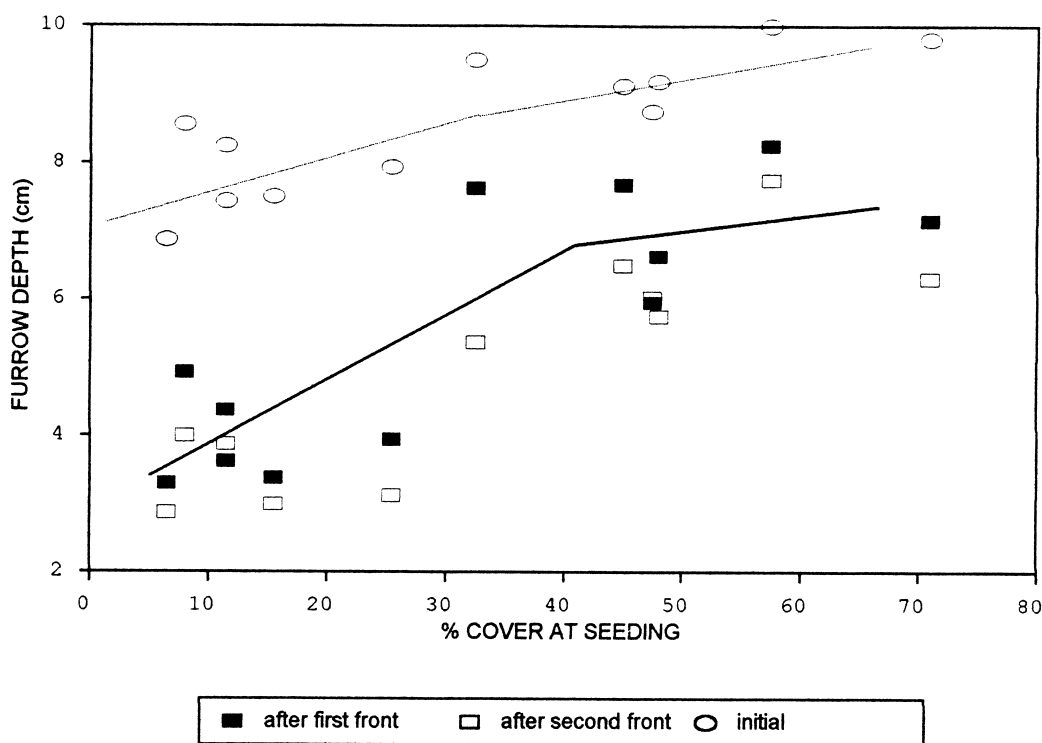


Fig. 5. Changes of furrow depth for different amounts of surface cover following the passage of two frontal systems at Badgingarra, 1994. Lines have been drawn through the data by eye. Loss of furrow depth is similar to the increased depth of seed. From Blackwell et al. (1994a).

8. Preferential surface flow of water

Field experiments were done on a dry water repellent sand near Geraldton, MED of 2.8, coarse sand 88%, fine sand 8.5%, silt 1.5%, clay 2% and total carbon content of 0.91%. A rainfall simulator and natural rain were used for precipitation on initially air-dry sand. The precipitation was on an artificial and simplified furrow of two flat sides, with different slopes and surfaces shedding water into a collection trough with instrumentation to measure runoff. Measurements were also made of soil wetting, by excavation and point intercept methods, on real furrows with or without narrow bands of surfactant in the furrow base. Further details are provided in Blackwell et al. (1994b).

The results of soil wetting below real furrows and the effect of different amounts of simulated rain (50 mm/h) and presence of different amounts of surfactant are shown in Fig. 3. The percentage interception of wet soil at seed depth (40 mm below the

furrow base or 40 mm below a level surface) increased as the amount of rain applied increased. A natural rainfall of 12 mm at 67 mm/h on the same furrows and adjacent level dry soil showed a very large increase in amount of wetted soil at seed depth, about 70% more wetted area. Addition of banded surfactant for the same amount of artificial rain increased the amount of wetted soil. Application of surfactant equivalent to 1 L/ha, uniform application, resulted in as much soil wetting as more than twice as much rainfall (2.5 mm compared to about 6 mm).

Measurements of runoff from the simplified artificial furrow showed slope has little effect on the amount of harvested water compared to furrow width and the presence or absence of prostrate or erect stubble.

Fig. 4 shows the output from a simple model derived from the results. No water is harvested until the surface layer and residue has been wetted, approximately 0.5–1 mm of rain. Water harvesting

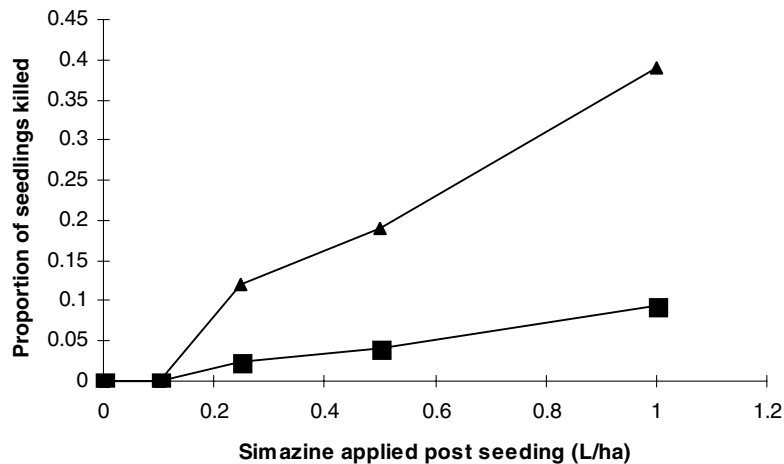


Fig. 6. Lupin seedling mortality from different rates of pesticide with (■) or without (▲) the protection of a wheat stubble.

is more efficient from a bare surface than with surface residues and an erect stubble restricts harvesting less than prostrate stubble.

This is a convenient synergism of stubble with agronomic needs. Water repellent soils cause more splash than non-repellent soils (Terry and Shakesby, 1993). Splashed soil can carry fungal spores and pesticides onto the plant, but erect stubble reduces the splash and makes water harvesting more efficient if there is surface residue. Larger catchments from wider furrows in Fig. 4 result in more water harvesting.

9. Preferential flow of soil

Furrow filling can result in seed deeper than ideal, but is considerably reduced by retained cover and reduced disturbance. Evidence of this is shown from measurements at Badgingarra in 1994.

Repeated measurements of furrow depth were made on dry furrow sown water repellent sand at Badgingarra before and after the strong wind and autumn rain in May 1994. Surface cover (vegetation and clods) at the same locations as measurements of furrow depth, was measured by point intercept methods. Rain and wind from initial fronts in May 1994 caused more furrow filling as surface cover of pasture residues and soil clods declined (Fig. 5). These results emphasise the need for adequate cover to minimise furrow

filling. Furrow filling can induce seed being buried too deep for successful emergence. Management methods need to conserve adequate surface covers, especially of pastures, and use sowing along contours where the risk of water erosion is increased by surface slope. Soil clods strengthened by dead plant roots are very valuable to provide protection in pastures, yet are easily lost by cultivation.

Soil carried into the bottom of a furrow can also concentrate associated pesticides in the crop row, leading to possible overdose of the crop and facilitating possible leaching through the bottom of the furrow.

10. Risks of pesticide concentration

Evidence of problems from pesticide concentration in furrows is shown from an experiment in 1993. Lupins were sown dry on 300 mm spaced furrows, 50 mm deep either in a pasture area or in stubble from a crop of wheat which yielded 2 t/ha the previous year. The sowing operation resulted in bare ridges from the pasture and ridges covered by 30% crop residue in the stubble. The crop was germinated by water harvested by a 9 mm shower in late April. The pesticide simazine was then sprayed on dry soil at different rates. A further shower of 9 mm occurred when the crop was emerging. Mortality of the Lupins (Fig. 6) increased rapidly for the bare ridges, killing

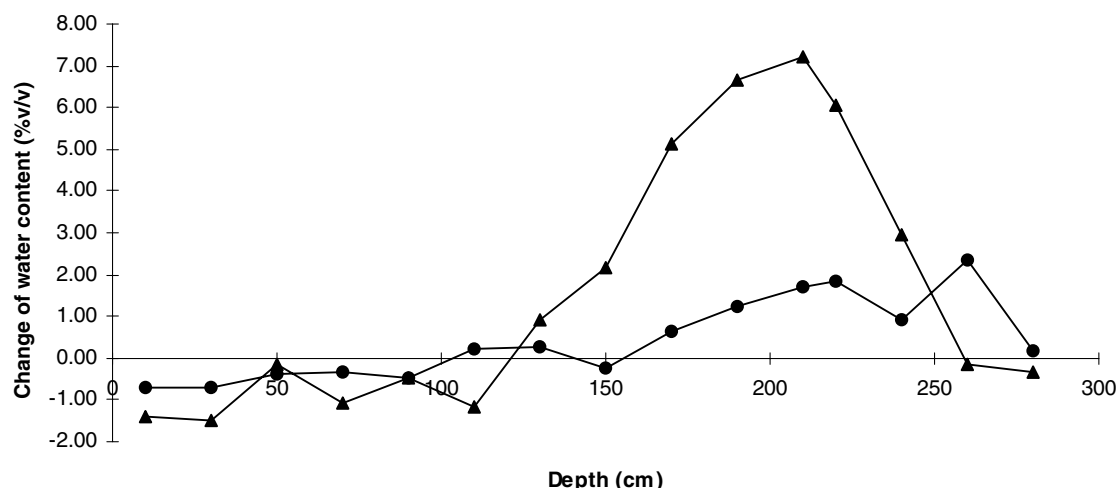


Fig. 7. The effect of wide furrows on profile water content in a duplex soil after 58 mm of natural rain (●) or 100 mm of artificial rain (▲). The texture change from sand to sandy clay is at 110 cm. The water content difference between paired sites with or without furrows, in percentage by volume, is shown at different depths. For the natural rain it is a mean of changes at four sites; the data between 150 and 240 cm depth shows significant changes at $P < 0.05$.

almost half the plants at the higher rate. The stubble residue reduced the mortality by more than 40%, because less soil and pesticide was moved into the bottom of the furrow, avoiding as much pesticide overdose of the crop.

This is an extreme example of pesticide toxicity due to increased dosage by diverted flow during water harvesting. Current practices minimise the problem by later application of broadleaf pesticides and formulations more easily metabolised by the lupin crop. Smaller furrows also help to minimise the risk.

11. Risks of leaching

A preliminary study of deep-water movement from furrows was made at Geraldton in 1992. Changes of water content beneath level or furrow sown crop row was evaluated with four paired sites, 3 m apart. At each paired site water content was measured with a neutron moisture meter through a 3 m deep pvc access tube. One tube was installed in a level crop row and one tube in a wide furrow sown crop row (355 mm spacing, 50 mm deep). The tubes were installed directly into the duplex profile using an internal auger, with suction to remove the debris, and the bottoms sealed by underwater cement. Regular

measurements were made through the 1992-growing season. The most information (Fig. 7) came from measurements in July before or after a 58 mm rain, the subsoil was significantly more wet beneath the furrows than beneath the level surface. The simulator applied 100 mm of rain and was controlled to avoid surface ponding in the furrow to minimise the risk of preferential flow along the interface between the outside of the access tube and the soil. The water content profiles after artificial rain was only successfully measured on two paired plots. The resulting changes of water content were so dissimilar between the pairs that only one profile is shown in Fig. 7. If this is still representative of the process, it shows a large influx of water into the B horizon by preferential water movement through the furrow. The pattern of wet soil below wide furrows revealed by excavation after rain has often shown the soil beneath the row to be much wetter than the soil beneath the ridge at the same depth. Thus wide furrows may be contributing to significantly more leaching of nutrients than level sowings and narrower furrows during heavy rains of about 100 mm or more. This hypothesis is supported by observations of wide furrow sown cereals at some farms in 1993 after 80 mm opening rains. The early growth of these cereals appeared less and the plants looked more nitrogen deficient than adjacent level

Table 1
An approximation of relative risk and hazard for increased groundwater contamination, from different systems

System	Relative risk	Relative area in use (WA 1998)	Relative use of pesticides and fertilisers	Actions to reduce risk	Relative current hazard (risk × area × pesticide use)	Relative potential hazard (risk × pesticide use)
Furrow sowing	Very high 4	Large 4	High 5	Smaller furrows more residue isolation in dry core	Very high 80	20
FS with banded surfactant	High 3	Small 1	High 5	Low persistence surfactants	Small 15	15
Zero till	Medium 2	Small 2	High 5	Maximise stubble retention	Moderate 20	10
Fodder shrubs	Medium 2	Moderate 3	Small 3	Fill trenches after establishment	Moderate 18	6
Tree plantations	Medium 2	Moderate 3	Small-high 3-5	Reduce pesticide use	Moderate-high 18-30	6-10
Blue Lupins	Very high 4	Small 2	Very small 1	Top dress fertilisers in winter	Very low 4	4
Claying	Low 1	Small 2	High 5	Deeper incorporation and higher quantities	Small 10	5
Microbial remediation	Low 1	Nil	High 5	Nil	Nil	5

sown crops. Thus there is a reasonable case for more detailed studies of leaching risk and effects on the water balance by wide furrows. This may be more of an advantage in lower rainfall areas and more of a problem in high rainfall areas. Management strategies, such as tactical applications of fertiliser to cereals, may be needed to correct any effects of severe nutrient leaching on crops sown in wide furrows.

These results of measurements of preferential flow of water, soil and pesticides in furrow systems emphasize the importance of surface cover, especially erect stubble and intact clods of pasture, and furrow width. Narrower furrows with most retention of cover will offer the least risk from problems caused by preferential flow. Such small furrows can be made more efficient for water harvesting by the addition of a surfactant band.

12. Relative risk and hazard of increased groundwater contamination

An estimate of risks and hazards of increased groundwater contamination, from different systems for managing repellency, can be made from existing information, experience and some assumptions. In Table 1, the relative risk is scored by the possible occurrence of preferential flow, thus furrow sowing is high and claying is low. Surfactants reduce the risk by extra soil wetting. Zero tilled soil still has risk, because it is still repellent, despite extra cover to reduce drying. Removal of repellency by masking (claying) or degradation has the most effect.

The relative use of pesticides and fertilisers combines with the risk and area in use to calculate a relative current hazard. A relative potential hazard assumes the systems could be used anywhere. Thus, furrow sowing has the greatest current and potential hazard and blue lupin pasture the least potential hazard, because no pesticides are used.

Possible actions to reduce risk are indicated in Table 1. Low persistence surfactants minimise the risk of reduced crop yield, despite improved establishment (Blackwell, 1997). Deeper incorporation of larger amounts of clay may improve solute adsorption and retention. Filling deep trenches after fodder trees are established may reduce the amount of

excess water harvesting. Reduction of pesticide use in tree plantations may reduce the overall hazard. Top-dressing fertiliser in winter, instead of summer or autumn, may reduce the hazard of nutrient leaching.

A better quantification of these risks and hazards needs further research using quantitative modelling. Quantitative modelling solutions for water repellent soils have already been useful to predict soil temperature and evaporation for different designs of furrow systems (Yang et al., 1996) and for changes in soil water content (Ritsema et al., 1998).

13. Conclusions

Management systems to reduce problems from water repellent soils in Australia have a wide range of relative risk of preferential flow, pesticide concentration and leaching of pesticide and nutrient; resulting in possible increased groundwater contamination. To help assess this risk, this analysis has used principles and processes involved in repellent soil and improved agricultural methods. Repellency problems can be simplified as either preferential flow or isolated volumes of dry soil. Repellency solutions can be classified as adaptation, avoidance, water harvesting, reduced surface tension, reduced drying, masking, degradation or export. Masking by claying and degradation offer the lowest increase of risk of contamination, because repellency is removed. Preferential flow through furrows offer the highest risks of increased contamination, and is mainly controlled by furrow width and soil cover. The narrowest furrows with the most cover offer the least risk; this approximates to a zero till system. Banded surfactant can make water harvesting in zero till more efficient. Banded surfactant can also allow more soil to be wetted, to adsorb more solute. Dead root systems in zero till may play a useful role and more research is needed on the effect of dead root systems on water redistribution and interaction between solutes, organic matter and microbial activity. The hazard depends on the amount of pesticide and fertiliser used, thus cropping systems are generally more at risk than grazing systems. Claying in combination with zero till offer potential long term benefits from increased adsorption and organic matter, however claying is a slow process and not feasible for all areas. Therefore there is an urgent

need for new research to enable good quantification of the risks. This could assist the formulation of management guidelines, tools and policies to identify the situations of greatest risk and reduce any possible hazards. Best current advice may be to minimise application of atrazine to dry water repellent soil in cropping systems using furrows.

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